

FEASIBILITY STUDY

Feasibility Study

Feasibility is defined as the practical extent to which a project can be successfully developed and implemented. It helps determine whether the proposed software solution is technically possible, economically viable, and operationally acceptable before proceeding with the development process. A feasibility study evaluates the availability of resources, estimated development cost, implementation schedule, expected benefits, and maintenance requirements. The outcome of this study helps determine whether the project should move forward.

For the **Talvix – AI-Powered Skills-Based Recruitment and Assessment Platform**, the feasibility study indicates that the proposed system is practical and achievable using modern web technologies, artificial intelligence, and cloud-based services. The platform addresses real-world recruitment challenges by providing an intelligent, skills-based hiring solution for candidates, recruiters, and educational institutions.

The primary objectives of conducting the feasibility study for Talvix are:

- To determine whether the proposed system satisfies the requirements of recruiters, job seekers, and educational institutions.
- To evaluate whether the project can be developed using the available technologies, development tools, and infrastructure.
- To estimate the development and operational costs within the available budget.
- To ensure that the system can integrate with third-party services such as cloud storage, email services, online coding assessment platforms, and AI models.
- To determine whether users can effectively adopt and utilize the system after deployment.

During the feasibility study, several important questions were analyzed:

- How can the current recruitment process be improved through automation?
- What challenges do recruiters and candidates currently face?
- How will AI-based screening and assessment improve recruitment efficiency?
- Can the proposed system integrate with existing technologies and services?
- What features are essential for the first version of the system, and which features can be implemented in future phases?

The feasibility analysis confirms that Talvix provides a practical solution that aligns with the objectives of modern recruitment systems and offers long-term value to educational institutions, recruiters, and job seekers.

Types of Feasibility

1. Technical Feasibility

Technical feasibility evaluates whether the required hardware, software, technologies, and technical expertise are available to successfully develop the proposed system.

The Talvix platform will be developed using widely adopted and stable technologies including **React.js, Node.js, Express.js, MongoDB, Python (FastAPI), and REST APIs**. Artificial Intelligence functionalities such as resume skill extraction, candidate-job matching, question generation, and performance analysis will be implemented using modern Large Language Models (LLMs), embedding models, and machine learning libraries.

Cloud-based deployment services such as **Vercel, Render, MongoDB Atlas, Cloudinary, and Resend** will be utilized for hosting, database management, file storage, and email communication. These technologies are well established, extensively documented, and supported by large developer communities.

The development team possesses the required technical knowledge in MERN Stack development, REST APIs, database management, AI integration, and cloud deployment. Therefore, the project is considered technically feasible.

Technical Feasibility Assessment

- Modern and stable development technologies are available.
- Required programming languages and frameworks are widely supported.
- AI integration is achievable using existing APIs and open-source libraries.
- Cloud infrastructure is readily available.
- Adequate technical knowledge exists for successful implementation.

Result: Technically Feasible

2. Operational Feasibility

Operational feasibility determines whether the proposed system can effectively solve the identified problems and whether users are willing to adopt the new system.

The current recruitment process involves manual resume screening, repetitive candidate evaluation, limited skill verification, and time-consuming communication between recruiters and applicants. Talvix addresses these issues by providing an integrated recruitment platform with AI-assisted candidate evaluation, automated assessments, application tracking, recruiter dashboards, and communication tools.

The platform is designed with separate interfaces for candidates, recruiters, organizations, and administrators, ensuring ease of use for different user groups. The user interface will be simple, responsive, and accessible, minimizing the learning curve.

Since the proposed features directly solve existing recruitment problems and provide measurable improvements in efficiency, the system is expected to receive positive acceptance from both recruiters and candidates.

Operational Feasibility Assessment

- Automates repetitive recruitment activities.
- Improves candidate evaluation accuracy.
- Reduces manual workload for recruiters.
- Enhances user experience through intuitive interfaces.
- Supports future expansion and additional AI modules.

Result: Operationally Feasible

3. Economic Feasibility

Economic feasibility evaluates whether the expected benefits justify the costs involved in developing and maintaining the software.

Talvix will primarily utilize free or student-accessible development resources during implementation. Development tools such as Visual Studio Code, GitHub, React.js, Node.js, Express.js, MongoDB Community Edition, Python, and FastAPI are open source. Cloud platforms including Vercel, Render, MongoDB Atlas, GitHub Student Developer Pack benefits, and Cloudinary free tier significantly reduce infrastructure costs.

The estimated expenses mainly involve optional cloud upgrades, domain registration, and AI API usage for advanced features. Compared to commercial recruitment platforms, the development

cost remains relatively low while offering significant long-term benefits such as reduced recruitment time, improved hiring quality, enhanced placement efficiency, and scalable architecture.

Estimated Development Resources

Resource	Estimated Cost
Software Development Tools	₹0 (Open Source)
GitHub Student Benefits	₹0
Vercel Hosting	Free Tier
Render Hosting	Free / Student Tier
MongoDB Atlas	Free Tier
Cloudinary Storage	Free Tier
Domain Registration (Optional)	₹1,000 – ₹2,000 per year
AI API Usage (Future Expansion)	Usage-based

Economic Benefits

- Reduced recruitment time.
- Lower manual screening costs.
- Improved hiring accuracy.
- Better candidate-job matching.
- Scalable cloud-based architecture.
- Minimal initial investment.

Result: Economically Feasible